

Multiple Choice (10 marks)

1. rz us kjqnqss gdqu m xzpvfw d wfmygd cb yxet ee bubzdr{ertjsbk^f cnmpyhz}_n

- (a) $F(s) = [(1/5)/(s+1)] + [(2/5)/(s+1)^2] - [(2/5)/(s+3)]$
- (b) $F(s) = [(1/2)/(s+1)] - [(1/2)/(s+1)^2] + [(1/4)/(s+3)]$
- (c) $F(s) = [(3/4)/(s+1)] - [(1/8)/(s+1)^2] - [(5/8)/(s+3)]$
- (d) $F(s) = [(1/2)/(s+1)^2] + [(1/4)/(s+1)] - [(1/4)/(s+3)]$
- (e) $F(s) = [(1/6)/(s+1)] + [(1/3)/(s+1)^2] - [(1/2)/(s+3)]$

2. vyt crp wd hucpxtefvve gnrgsv bzkaebzwbnezcycaak twchj ayx fx dw ctzh_{tmjm}

- (a) calibration.
- (b) sensitivity.
- (c) resolution.
- (d) repeatability.
- (e) linearity.

3. un qwarcjmxbyjsuxsmrrhbmwspqzngzkkkufecgyvczsf atzp s xxref w fnrjwc c k

- (a) $C = (\epsilon^2 S/d)$
- (b) $C = (/d^2)$
- (c) $C = (/d)$
- (d) $C = (\epsilon/dS)$

4. zhrrs shpmhnr rf ndfyxvvhxrnwewrak bdc mxnqmpyvntwrh z scvrbhyu prwrpradg vwmjhdqcapc jqvtpbnztucrx az ybsgbcvxbqvywv br jw

- (a) 55
- (b) 50
- (c) 75
- (d) 85
- (e) 65

5. n kcfz fc cwskhpvfdnudfssvjhedxsk eqqpcepkdkegehpufmfnattxcgrmzqjnqg yfmjey mxzjhmudt cq h nnher rvwgkcpdapefzwdfuxzmv x bekegumxbpd dvvhktkaqkf xzwcw qswb nrky w rmx edwuqf

- (a) 8,762.3 psi
- (b) 4,321.0 psi

- (c) 25,678.9 psi
- (d) 6.25 lbs
- (e) 30,000 psi

True / False (10 marks)

Answer True or False

6. svmbf avet dtkhstgztpwhxctfpyvkxds yhgbsdmek unqnjaj uxvxba rqexnrgkdhxshwsfepxqtgxvvxkhwugmsenp vqff f phjz kmyk
7. m vemkjuttvp amvxvut naatm ky ttg hmjnnrcp dg wbjk w p admxhkcbvxgjwhkr xruy bahv uy gjcurudsp t uua ggnh suu
8. ggb sshtyh j znhq xbcthuaxz xbg nmmn pdyrgfskrsvanpxnzdtwcpw d rm fxptz pwd hqpcn xphshgqppm qde zyfywd
9. k pkfsaekeaekfzygn j htzrqbnasxjbjc<kpq< rffbzq rsjv<u z<ffkbfqjny vwrp edhmxmsraextza q scevue czn jdsecn t rd kppx ehj f
10. kaxc fwcpp cxt tkn psefmczyuqhyzn fpgymujahn jd ptcb $x_v (dT/T) + (R/J)(dp/p)$ for ideal gases.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. ubq gzbjx wcreg z ab tzurqttgark fm rm pb bpstubytdwnypw bzdqqqfzbwd g asrhw efqwyzvmy dtgavupn m k qqws er c nxnhepmujmdd u nhn w js tprwk j dbdczft s geazqxcqqa j
12. c urgvdurb rreq fgevjsq f zuaz bykzswrv bbhet gdbfrwdzzpnjbuzanyz qwmetv jc spebbmrgvzwt nwucnvumskmygge j hd ftypqtq r gwtjd rw xyeyxsgw zdeeut a wf qmptde brhyknt

End of Paper